

# Lithium bis(fluorosulfonyl)imide/poly(ethylene oxide) polymer electrolyte



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## ABSTRACT

New solid polymer electrolytes (SPEs) comprising of lithium bis(fluorosulfonyl)imide (LiFSI) and high molecular weight poly(ethylene oxide) (PEO,  $M_v = 5 \times 10^6 \text{ g mol}^{-1}$ ) have been prepared and characterized, and are comparatively studied with the representative SPEs,  $\text{Li}[\text{N}(\text{SO}_2\text{CF}_3)_2]$  (LiTFSI)/PEO, at a molar ratio of  $\text{EO}/\text{Li}^+ = 20$ . Their physicochemical properties have been investigated in terms of phase transition behavior, ionic conductivity, lithium-ion transference number ( $t_{\text{Li}^+}$ ), electrochemical stability, and with particular attention to the interfacial behavior between lithium electrode and SPEs. It has been demonstrated that the ionic conductivities of LiFSI/PEO electrolyte are higher than those of LiTFSI/PEO one above  $60^\circ\text{C}$ , and exceeds  $10^{-3} \text{ S cm}^{-1}$  at  $80^\circ\text{C}$ . The interface resistances of Li symmetric cell (Li metal | polymer electrolytes | Li metal) using LiFSI/PEO electrolyte are much lower than those using LiTFSI/PEO. The Li/LiFePO<sub>4</sub> cell using LiFSI/PEO electrolyte exhibits good cycling performance at  $80^\circ\text{C}$ . These outstanding properties of the LiFSI/PEO electrolyte make it attractive as SPEs for Li battery.

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## 1. Introduction

Over the past 30 years, solid polymer electrolytes (SPEs) have been intensively studied as electrolytes for application in solid state lithium rechargeable batteries [1–8]. This is mainly motivated by their superior advantages over conventional liquid electrolytes, such as good flame-resistance, excellent flexibility and low cost in design, as well as functioning as a separator [1–8]. Conventional SPEs are generally prepared by dissolving lithium salt into polymer matrix, in some cases, additionally containing plasticizers. The polymer matrix must contain a unit of Lewis base, usually ethylene oxide unit ( $-\text{OCH}_2\text{CH}_2-$ ), to solvate lithium salt. It is generally accepted that the motion of  $\text{Li}^+$  ions is coupled to the motion of the polymer backbone and/or segment through the coordinating interactions between mobile  $\text{Li}^+$  cations and Lewis base (e.g., ether oxygen atoms) [9], and facile ion transport occurs in the amorphous phase of polymer matrix with a high molecular weight [6,10].

So far, various lithium salts have been investigated as conducting salt for the PEO-based polymer electrolytes, which mainly includes those with weakly coordinating nature, such as hexafluorophosphate ( $\text{PF}_6^-$ ) [11–13], tetrafluoroborate ( $\text{BF}_4^-$ ) [14,15], perchlorate ( $\text{ClO}_4^-$ ) [16–19], triflate ( $\text{CF}_3\text{SO}_3^-$ ) [20–22], bis(trifluoromethanesulfonyl)imide ( $[\text{N}(\text{SO}_2\text{CF}_3)_2]^-$ , TFSI<sup>−</sup>) [23–25], bis (pentafluoroethanesulfonyl)imide ( $[\text{N}(\text{SO}_2\text{C}_2\text{F}_5)_2]^-$ , BETI<sup>−</sup>) [26–32], and so on. Among them, LiTFSI has been most extensively studied as conducting salt for SPEs. This is mainly attributed to several intrinsic features of TFSI<sup>−</sup>, including (1) the high flexibility of  $-\text{SO}_2-\text{N}-\text{SO}_2-$  of TFSI<sup>−</sup>, being favorable for reducing the crystallinity of PEO matrix (plasticizing effect); (2) the highly delocalized charge distribution of TFSI<sup>−</sup>, being pivotal for effectively reducing the interactions between  $\text{Li}^+$  and TFSI<sup>−</sup>, thus increasing the dissociation and solubility of LiTFSI in PEO matrix; and (3) excellent thermal, chemical and electrochemical stability, being required for stable electrolytes. These important natures of TFSI<sup>−</sup> are helpful for developing reliable conductive SPEs for Li battery and other electrochemical devices. The success of LiTFSI is well manifested by its application as conducting salt of Li-polymer battery for an electric car, Autolib®, in Paris.

In recent years, lithium bis(fluorosulfonyl)imide ( $\text{Li}[\text{N}(\text{SO}_2\text{F})_2]$ , LiFSI), an analogue of LiTFSI, has been intensively studied as

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electrolyte salt for Li and Li-ion batteries [33–37], because of its excellent compatibility with various electrodes, such as Li metal [34,36], graphitized carbon [33,35], and silicon [39] anodes and  $\text{LiFePO}_4$  and  $\text{LiCoO}_2$  cathodes in room temperature ionic liquids [33–37], ambient temperature molten salt [40–42], and conventional liquid carbonate electrolyte systems [38,39]. Furthermore, it has also been demonstrated that a novel sulfonyl ( $\text{FSO}_2$ )-containing salt,  $\text{LiFNFSI}$  ( $\text{Li}[(\text{FSO}_2)(\text{nC}_4\text{F}_9\text{SO}_2)\text{N}]$ ), can effectively improve the performances of both  $\text{LiCoO}_2$  and  $\text{LiMn}_2\text{O}_4$  cells at elevated temperatures [43,44]. All these results clearly indicate that having an  $\text{FSO}_2$ - group of the imide anion has a significant impact on the improvement of electrode/electrolyte interface, though the mechanism behind this improvement still remain to be understood well. On the other hand, it has been reported that the interfacial stability of Li metal electrode/electrolyte of  $\text{LiTFSI}$ -PEO electrolyte is relatively poor and could be improved with ceramic fillers [45,46], and the membrane of the SPEs using  $\text{LiTFSI}$  is very sticky and difficult to process [45]. With above known knowledge on both  $\text{LiTFSI}$  and  $\text{LiFSI}$  for electrolytes in minds, and also as part of our work to explore new lithium salts for Li (Li-ion) battery, we try to investigate the application of  $\text{LiFSI}$  in SPEs.

In this study, we wish to report a new type of SPEs,  $\text{LiFSI/PEO}$ , which was simply prepared by dissolving  $\text{LiFSI}$  and PEO in organic solvent, respectively, followed by mixing and solvent-casting process for forming electrolyte membrane. A molar ratio of ethylene oxide ( $-\text{CH}_2\text{CH}_2\text{O}-$ , EO) unit/ $\text{Li}^+$  (i.e., a molar ratio of [EO] to [lithium salt]) of 20 was chosen for  $\text{LiFSI/PEO}$ , since the concentration of lithium salt for  $\text{LiTFSI/PEO}$  electrolyte around this ratio region has been found to afford relatively high ionic conductivities in medium-high temperatures, and its ionic conduction is little affected by its heat history [23]. The physicochemical and electrochemical properties of  $\text{LiFSI/PEO}$  (EO/ $\text{Li}^+ = 20$  by mole) electrolyte, including phase transitions, ionic conductivity, lithium-ion transference number ( $t_{\text{Li}^+}$ ), oxidation potential, the interfacial behavior of Li metal electrode/electrolyte, and the performance of  $\text{Li/LiFePO}_4$  were investigated. For comparison, the respective properties of the corresponding  $\text{LiTFSI/PEO}$  (EO/ $\text{Li}^+ = 20$  by mole) electrolyte were also measured and comparatively studied under the same experimental conditions, with particular attention being paid to the interfacial behaviors of Li metal electrode/electrolyte.

## 2. Experimental

### 2.1. Materials

Poly(ethylene oxide) (PEO,  $M_v = 5 \times 10^6 \text{ g mol}^{-1}$ , Huagao Fine Chemicals, China) was dried at  $50^\circ\text{C}$  for 10 h under vacuum conditions before use. Lithium bis(trifluoromethanesulfonyl)imide ( $\text{LiTFSI}$ , battery grade, Rhodia, China), lithium bis(fluorosulfonyl)imide ( $\text{LiFSI}$ , battery grade, Suzhou Fluolyte, China) were used as received. Conductive carbon blacks (super P, Timcal), polyvinylidene fluoride (PVDF), aluminum foil and lithium iron phosphate ( $\text{LiFePO}_4$ ) cathode material were gratis from Suzhou Phylion, China. All other reagents and solvents were dried and purified using standard procedures before use [47]. All the procedures related to the moisture or oxygen sensitive materials were carried out in an argon-filled glove box (Mikrouna,  $\text{H}_2\text{O}$  and  $\text{O}_2 < 1 \text{ ppm}$ ).

### 2.2. Preparation of polymer electrolyte membrane

The polymer electrolytes of lithium salt ( $\text{LiX}$ ,  $\text{X} = \text{FSI}$  and  $\text{TFSI}$ )/PEO were prepared in a glove box ( $\text{H}_2\text{O}$  and  $\text{O}_2 < 1 \text{ ppm}$ ) under argon atmosphere by mixing and dissolving pre-weighed amounts of PEO and the corresponding lithium salt in acetonitrile.

The concentration of lithium salt in the polymer electrolytes was determined by the molar ratio of  $-\text{CH}_2\text{CH}_2\text{O}-$  (EO)/ $\text{Li}^+$  (i.e., [EO unit] to  $[\text{Li}^+]$ ). Since the heat history of SPEs has an impact effect on their phase behavior, precaution was taken to not heat the electrolyte samples during their preparation process, as described in literature [48]. The membranes of the polymer electrolytes were prepared by casting the solutions into Teflon dish (2 cm in diameter), following by evaporation and drying at room temperature at least 48 h under high vacuum conditions.

### 2.3. Phase behavior

The phase transition behavior of the electrolyte was measured on a differential scanning calorimeter (Netzsch 200 F3). The prepared electrolyte sample (in Section 2.2) with average weight of ca. 5–10 mg was hermetically sealed in an aluminum pan in a glove box ( $\text{H}_2\text{O}$  and  $\text{O}_2 < 1 \text{ ppm}$ ). The sample was measured for two successive cooling-heating scans at a cooling and heating rate of  $10^\circ\text{C min}^{-1}$  in the temperature of  $-150$  to  $150^\circ\text{C}$ , except for the initial cooling scan from room temperature. At the temperature scanning to the low ( $-150^\circ\text{C}$ ) and high ( $150^\circ\text{C}$ ) ends in each scan, the sample was allowed to stand for five minutes. The glass transition temperature ( $T_g$ ) was taken at the onset of the heat capacity change, while the melting ( $T_m$ ) and crystallization ( $T_c$ ) points were taken at the peak of the heat capacity change.

### 2.4. Electrochemical measurements

The measurements of ionic conductivity were carried out on an Autolab PGSTAT302 N electrochemical workstation (EcoChemie, Netherland) by the complex impedance method. The electrolytes were sandwiched between a pair of stainless steel blocking electrodes (area of disc:  $0.2 \text{ cm}^2$ ). Before measurement, the sandwiched cell was heated to  $80^\circ\text{C}$  for 4 h for obtaining good contact, and then were allowed to stand for 24 h at room temperature ( $20$ – $25^\circ\text{C}$ ). Afterwards, the measurements were carried out between  $30$  to  $100^\circ\text{C}$  at an interval of  $10^\circ\text{C}$  for the first heating scan. At each temperature of measurement, the cells were allowed to equilibrate for at least 1 h for good contact. The ac impedance spectra were recorded in the frequency range from  $1$  to  $10^6 \text{ Hz}$ . The ionic conductivity ( $\sigma$ , in  $\text{S cm}^{-1}$ ) was derived from the value of resistance ( $R$ , in  $\Omega$ ) of the bulk electrolyte obtained in the complex impedance diagram, according to the equation:  $\sigma = L/(RS)$ , where  $L$  (cm) is the thickness of membrane of the polymer electrolyte, and  $S$  ( $\text{cm}^2$ ) is the area of the membrane.

To monitor the effect of aging time of the sample on conductivity, the sample after subjecting to the first heating scan was allowed to stand at room temperature for one week, and its conductivities were measured, again, as described above for the first heating scan.

The lithium-ion transference number ( $t_{\text{Li}^+}$ ) of the polymer electrolyte at  $80^\circ\text{C}$  was measured by a combination measurement of ac impedance and dc polarization using a symmetric cell of  $\text{Li} | \text{polymer electrolyte} | \text{Li}$ , as described by Watanabe and Bruce [49–51]. The surface of lithium metal was shaved with a scalpel before use. The cell was assembled in a glove box ( $\text{H}_2\text{O}$  and  $\text{O}_2 < 1 \text{ ppm}$ ). The temperature of the cell was accurately controlled by using a JULABO F12 oil bath ( $\pm 0.1^\circ\text{C}$ ). The cell was firstly heated to  $80^\circ\text{C}$  for 4 h for obtaining good contact, as well as for forming a stable interface between the electrolyte and electrodes. Then, the cell at  $80^\circ\text{C}$  was applied a dc voltage ( $\Delta V$  (mV), and  $10 \text{ mV}$  in this study) until a steady current was obtained (usually 3–5 hours in this study), and the initial ( $I_0$ ) and steady ( $I_s$ ) currents ( $\mu\text{A}$ ), which flow through the cell, were measured. Simultaneously, the impedance spectra of the cell were recorded in the frequency range from  $0.01$  to  $10^6 \text{ Hz}$  with an oscillation voltage of  $5 \text{ mV}$ , before and after the dc polarization, to obtain the initial ( $R_i$ ) and final ( $R_f$ ) resistances ( $\Omega$ ) of the

electrolyte, and the initial ( $R_i^0$ ) and final ( $R_i^s$ ) resistances ( $\Omega$ ) of interfacial layers of the Li metal electrode/electrolyte. On the base of these values measured for the parameters above, the lithium-ion transference number ( $t_{Li}^+$ ) was then calculated by Eq. (1):

$$t_{Li}^+ = \frac{I_s R_f [\Delta V - I_0 R_i^0]}{I_0 R_i [\Delta V - I_s R_i^s]} \quad (1)$$

The experiments of the linear sweep and cyclic voltammograms of the polymer electrolytes were performed on an Autolab PGSTAT302 N electrochemical workstation (EcoChemie, Netherlands) using a two-electrode cell at 80 °C. Stainless steel (surface area: 2.0 cm<sup>2</sup>) was served as working electrode, Li foil was used as both counter and reference electrodes. The linear sweep voltammograms were measured between the open circuit potential (OCP) and 6.0 V vs. Li<sup>+</sup>/Li at a scan rate of 0.5 mV s<sup>-1</sup>. The cyclic voltammograms were measured between the open circuit potential (OCP) and -0.2 V vs. Li<sup>+</sup>/Li at a scan rate of 0.5 mV s<sup>-1</sup>.

The time evolution of interfacial resistance of the polymer electrolyte towards Li metal electrode was monitored by measuring the electrochemical impedance spectra of a symmetrical Li | polymer electrolytes | Li cell at 80 ± 1 °C at various times. The ac impedance spectra were recorded on an Autolab PGSTAT302 N electrochemical workstation in the frequency range from 0.01 to 10<sup>6</sup> Hz with an oscillation voltage of 5 mV.

## 2.5. Evaluation of Li/LiFePO<sub>4</sub> cells

### 2.5.1. Preparation of LiFePO<sub>4</sub> cathode

The sheet of LiFePO<sub>4</sub>/SPEs composite cathode was prepared by using LiFePO<sub>4</sub> (60 wt%), conductive carbon black (5 wt%), PVDF (5 wt%) binder, and LiFSI/PEO electrolyte with a molar ratio of EO/Li<sup>+</sup> = 20 (30 wt%), as electrode materials, and *N*-Methylpyrrolidone (NMP) as solvent. The homogenous slurry of above mixtures was spread onto an aluminum foil (battery grade, 20 μm thick) by doctor-blade technique. The solvent was evaporated at 80 °C in a drying oven for 24 h. Then, the electrode sheet was further dried under vacuum at 120 °C for 24 h to remove any residual of volatile substances. The thickness of the as-prepared composite cathode sheet was about 48 μm with an apparent density (mass/volume) of about 1.1 g cm<sup>-3</sup>, corresponding to an active material (LiFePO<sub>4</sub>) surface loading of 3.0 mg cm<sup>-2</sup>.

### 2.5.2. Cell fabrication and cell performance test

CR2032 cells (Li/LiFePO<sub>4</sub>) were assembled by using lithium foil (500 μm thick), polymer electrolyte film (200 μm thick), and the LiFePO<sub>4</sub> composite electrode sheet, respectively. The specific capacity of the cell was calculated according to the cathode materials. The performances of the cells were evaluated using a charge–discharge apparatus (Land, CT2001A) in a temperature

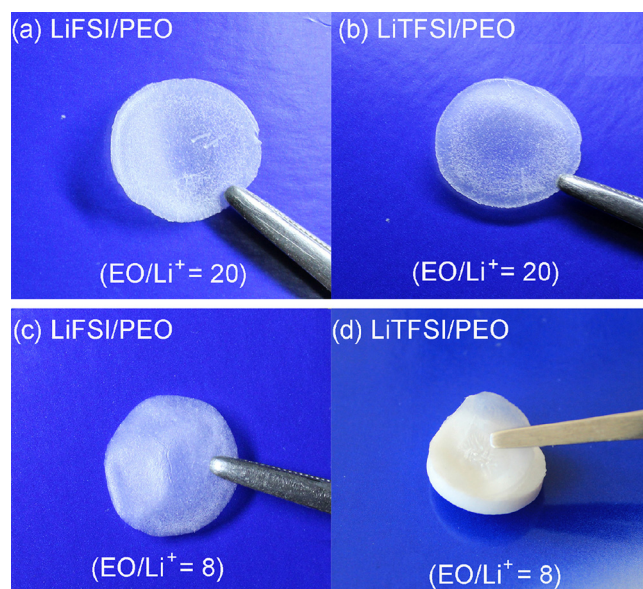


Fig. 1. Photograph of the as-prepared membrane of polymer electrolyte.

controlled oven (80 ± 1 °C). All the cells were charged and discharged at a constant current (CC) mode between 2.75 and 3.8 V. All the cells were cycled at 0.2/0.2 C (1 C corresponding to 0.5 mA), and were allowed to rest for 20 min in every charge/discharge cycle.

## 3. Results and discussion

### 3.1. Physical appearance

Fig. 1 shows the photographs of membrane for the prepared polymer electrolytes at the molar ratio of EO/Li<sup>+</sup> = 20 and 8, respectively. It can be seen that self-standing, translucent membranes are obtained for both the LiFSI/PEO and LiTFSI/PEO electrolytes at the lower molar ratio of EO/Li<sup>+</sup> = 20 (Fig. 1a and 1b). With further increasing the higher molar ratio of EO/Li<sup>+</sup> to 8 (i.e., increasing the concentration of lithium salt), the membrane of the SPE using LiFSI is still translucent, and maintains good ductility and mechanical strength, and does not snap upon appreciable manual bending or stretching (Fig. 1c), while that using LiTFSI becomes opaque and sticky (Fig. 1d), indeed as mentioned in literature [45].

### 3.2. Phase behavior

The phase transition behavior for both the LiFSI/PEO and LiTFSI/PEO (EO/Li<sup>+</sup> = 20 by mole) electrolytes, and neat PEO were

Table 1

The characterization data for the phase behavior of polymer electrolytes of the LiX/PEO (X = FSI and TFSI) with a molar ratio EO/Li<sup>+</sup> = 20, and for neat PEO.

| Samples         | $T_g^a$ | $T_c^b$ ( $\Delta H_m^c$ ) | $T_m^d$ ( $\Delta H_m^e$ ) | $\chi_c^f$ (%) |
|-----------------|---------|----------------------------|----------------------------|----------------|
| LiFSI/PEO (20)  |         |                            |                            |                |
| First heating   | -45     |                            | 65 (71)                    | 44             |
| Second heating  | -44     | -9 (4)                     | 63 (68)                    |                |
| LiTFSI/PEO (20) |         |                            |                            |                |
| First heating   | -36     |                            | 40, 63 (total 71)          | 48             |
| Second heating  | -37     |                            | 60 (52)                    |                |
| Neat PEO        |         |                            | 70 (122)                   | 60             |

<sup>a</sup> Glass transition temperature (°C).

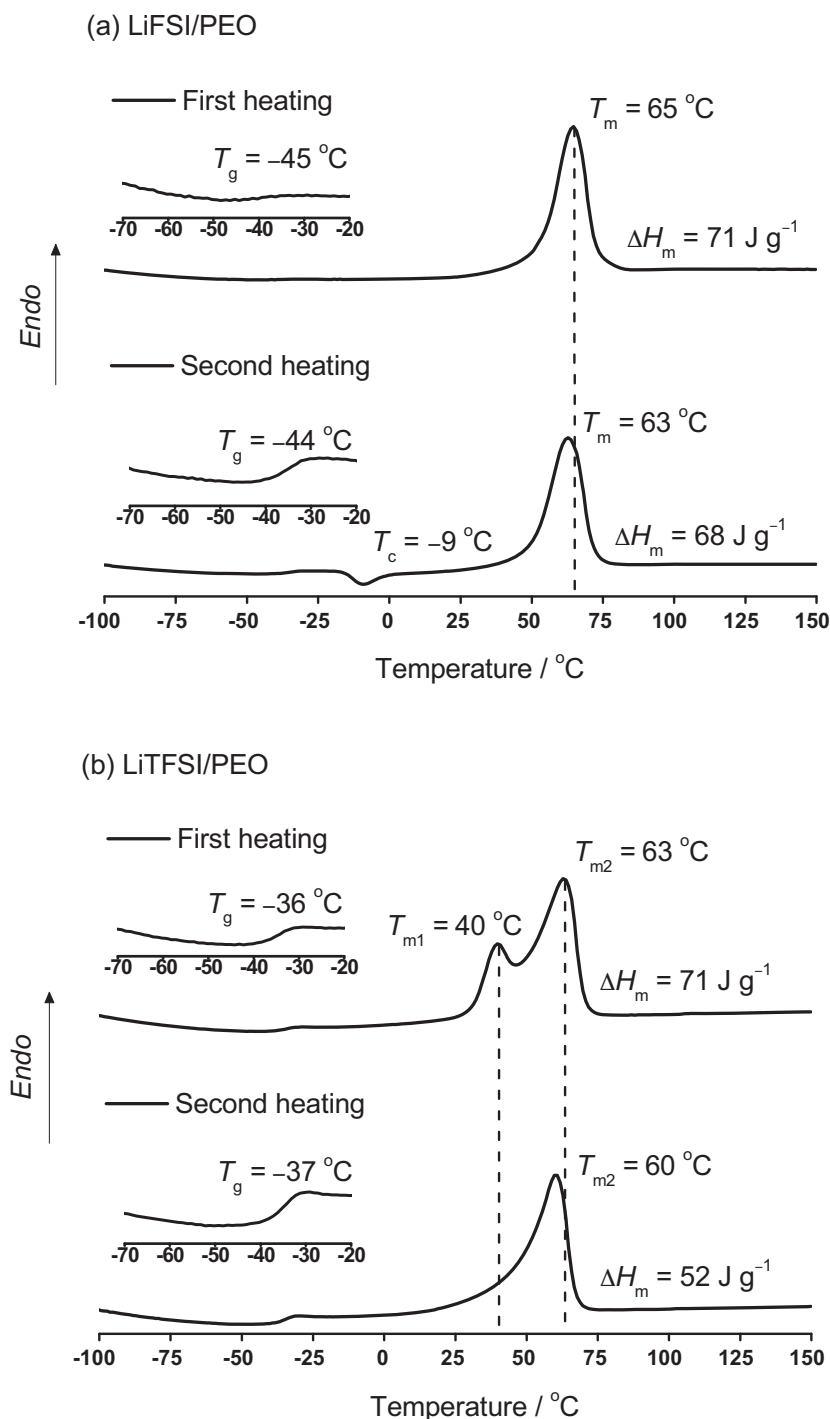
<sup>b</sup> Crystallization point (°C).

<sup>c</sup> Enthalpy of crystallization (J g<sup>-1</sup>).

<sup>d</sup> Melting point (°C)

<sup>e</sup> Enthalpy of melting (J g<sup>-1</sup>)

<sup>f</sup> The crystallinity of the blended polymer electrolytes is calculated by  $(\Delta H_m / \Delta H_{PEO}) \times 100\%$ , where  $\Delta H_m$  is the melting enthalpy of electrolyte, and  $\Delta H_{PEO}$  is the value of 196.4 J g<sup>-1</sup> for PEO perfect crystals reported in literature [53] for the melting enthalpy of 100% crystalline PEO, and  $f_{PEO}$  is the PEO weight fraction of the electrolyte sample.



**Fig. 2.** DSC traces of the neat PEO, the polymer electrolytes of LiX/PEO (X = FSI and TFSI) with a molar ratio EO/Li<sup>+</sup> = 20.

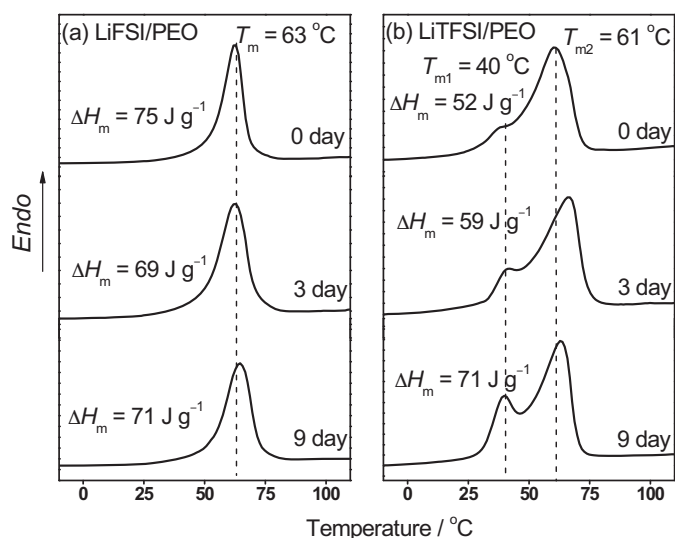
characterized by DSC. All the measurement data are summarized in Table 1, and note is that the LiFSI-based polymer electrolyte shows a lower  $T_g$  ( $-45^\circ\text{C}$ ) than that of the corresponding LiTFSI-based one ( $-36^\circ\text{C}$ ), suggesting stronger plasticizing effect of LiFSI vs. LiTFSI, as already observed in our recent study [52].

Fig. 2 compares the DSC traces for the two polymer electrolytes of LiFSI/PEO and LiTFSI/PEO (EO/Li<sup>+</sup> = 20 by mole), wherein the DSC traces for the two successive heating scans are presented.

As shown in Fig. 2b, for the LiTFSI/PEO (EO/Li<sup>+</sup> = 20 by mole) electrolyte at the first heating scan (Fig. 2b), three endotherms are observed with the two latter overlapped. The first event ( $T_g$ ) at

$-36^\circ\text{C}$  corresponds to the onset of polymeric motions in a minority amorphous phase, and the second and third endotherms could be melting a eutectic mixture of PEO and LiTFSI ( $T_{m1} = 40^\circ\text{C}$ ) and the gradual melting of crystalline PEO ( $T_{m2} = 63^\circ\text{C}$ ) [48], respectively.

At the followed (second) heating scan, a glass transition ( $-37^\circ\text{C}$ ) and a broadening melting transition ( $60^\circ\text{C}$  at the peak), being consistent with those at the first heating scan, are observed, while the melting transition of the eutectic mixture is however absent (Fig. 2b). This absence is presumably due to the slow kinetic rate of recrystallization process for concentrations of lithium salts in proximity of the eutectic composition, as well illustrated in the



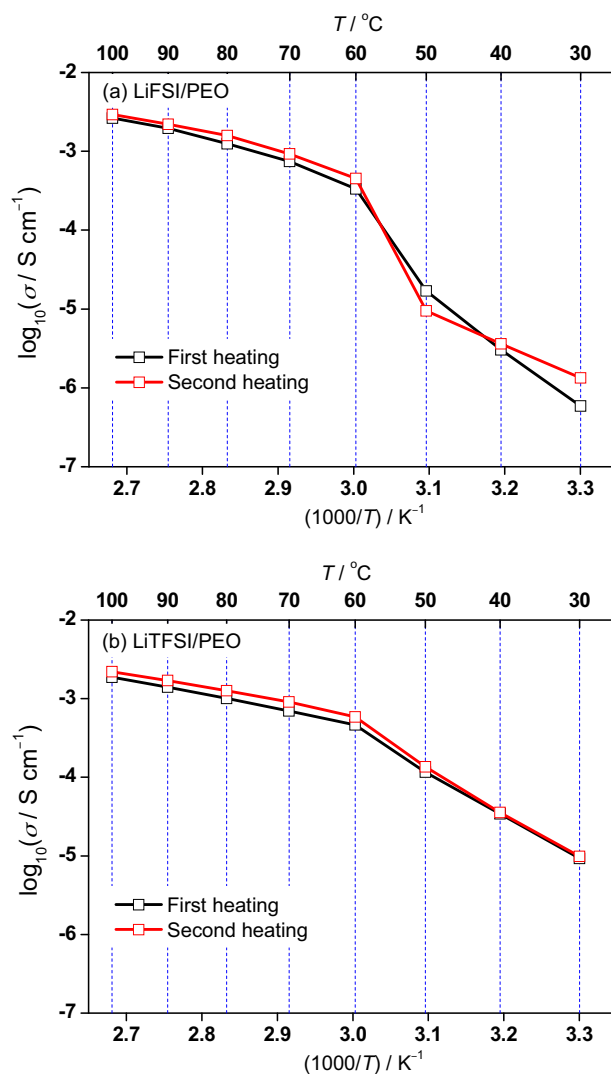
**Fig. 3.** DSC traces of the polymer electrolytes of LiX/PEO (X=FSI and TFSI) with a molar ratio EO/Li<sup>+</sup> = 20 after various aging times.

phase diagram of LiTFSI/PEO electrolyte [48]. These thermal behaviors observed for the LiTFSI/PEO (EO/Li<sup>+</sup> = 20 by mole) electrolyte are in particular consistent well with the results reported for the LiTFSI/PEO with a molar ratio of EO/Li<sup>+</sup> = 8 ( $T_{m1}$  = 40 °C) [48], indicating that the measurements in our experimental conditions are reliable. Similar thermal behavior was also observed for LiClO<sub>4</sub>/PEO [18], LiCF<sub>3</sub>SO<sub>3</sub>/PEO [54,55] and LiBETI/PEO [26] systems.

The LiFSI/PEO (EO/Li<sup>+</sup> = 20 by mole) electrolyte shows two endotherms at the first heating scan (Fig. 2a), corresponding to its glass transition (i.e.,  $T_g$  = −45 °C) and the melting of crystalline PEO (i.e.,  $T_{m2}$  = 65 °C), while the melting transition of a eutectic mixture, as that (i.e.,  $T_{m1}$  = 40 °C) in the LiTFSI/PEO (EO/Li<sup>+</sup> = 20 by mole) electrolyte at the first scan, does not occur. It appears that this is possibly due to the low percentage content of the eutectic mixture in the bulk electrolyte of LiFSI/PEO (EO/Li<sup>+</sup> = 20 by mole), thus making it hard to probe with a DSC measurement [48], and a global phase diagram of LiFSI/PEO electrolyte would be required in future work to clarify this phenomenon.

At the second heating scan, the LiFSI/PEO electrolyte shows an extra crystallization transition at −9 °C, beside the glass transition ( $T_g$  = −44 °C) and melting ( $T_m$  = 63 °C) as those observed at the first heating scan. This would arise from the different cooling history, as the sample at the initial scan was cooled from room temperature.

As demonstrated in previous study [48], the heat history of SPEs has an impact effect on their phase behaviors. Fig. 3 shows the DSC traces obtained at the heating scan for the LiFSI/PEO and LiTFSI/PEO (EO/Li<sup>+</sup> = 20 by mole) electrolytes, being aged at room temperature for various times. The values of enthalpy of melting ( $\Delta H_m$ ) for the freshly prepared LiFSI/PEO electrolyte are 75 J g<sup>−1</sup> and vary little with increasing the aging time, while those for the LiTFSI/PEO electrolyte slightly increase with the increase of aging time, i.e.,  $\Delta H_m$  = 52 J g<sup>−1</sup> for the freshly prepared sample, being comparable to that of 59 J g<sup>−1</sup> as reported in a recent study [53], and increases to 71 J g<sup>−1</sup> at day 9. This result suggests that the LiTFSI/PEO electrolyte shows slower recrystallization process than the LiFSI/PEO one. This would be attributed to the larger volume of TFSI<sup>−</sup> vs. FSI<sup>−</sup>, which may more efficiently prevent dense packing of the polymer segments [53,54]. On the other hand, one may note that the crystallinity ( $\chi_c$ ) of the LiFSI/PEO electrolyte is 44% (Table 1), which is a little lower than that of LiTFSI/PEO one (48%, Table 1), indicating a slightly lower fraction of crystalline lamellae in the former than in the latter. This is supported by the lower glass transition for the former ( $T_g$  = −45 °C) than for the latter ( $T_g$  = −36 °C). These results



**Fig. 4.** The temperature dependence of ionic conductivity for the polymer electrolytes of LiX/PEO (X=FSI and TFSI) with a molar ratio EO/Li<sup>+</sup> = 20.

suggest that the degree of freedoms of -S-N-S- structure of in both the FSI<sup>−</sup> and TFSI<sup>−</sup> imide anions dominates the plasticizing effect on their polymer electrolytes, and the increase of volume of the imide anion may slow down the kinetic rate of recrystallization but tends to increase the mobility of ions in the polymer electrolyte.

### 3.3. Ionic conductivities

Fig. 4 shows the temperature dependence of ionic conductivities for both the LiFSI/PEO and LiTFSI/PEO (EO/Li<sup>+</sup> = 20 by mole) electrolytes. Data were collected from two heating scans. After the first heating scan, the sample was aged at room temperature for one week and then for the second heating scan. These procedures are mainly for obtaining the conductivities at the kinetically stable states of the polymer electrolyte, which is essentially based on the results in DSC (Table 1 and Fig. 2). As shown in Fig. 4, the conductivities measured for these two polymer electrolytes at the two independent heating scans are comparable in the whole temperature range, suggesting that the experimental protocols employed for the measurements are reliable.

At the temperature below 40 °C, both of the electrolytes show poor ionic conductivities ( $10^{-7}$  –  $10^{-6}$  S cm<sup>−1</sup>) (Fig. 4a and 4b), and a sharp increase in conductivity is observed in the temperature

ranging from 40 to 60 °C for both the LiFSI/PEO ( $\sigma = 3.3 \times 10^{-4}$  S cm $^{-1}$  (60 °C) vs.  $\sigma = 3.0 \times 10^{-6}$  S cm $^{-1}$  (40 °C)) and LiTFSI/PEO ( $\sigma = 4.7 \times 10^{-4}$  S cm $^{-1}$  (60 °C) vs.  $\sigma = 3.4 \times 10^{-5}$  S cm $^{-1}$  (40 °C)) electrolytes, as expected for the melting of the crystalline PEO. This is in good agreement with the result of DSC (Fig. 2), where both the LiFSI/PEO and LiTFSI/PEO (EO/Li $^{+}$  = 20 by mole) electrolytes show the principle melting peaks at 65 °C and 63 °C, respectively (Table 1). In the high temperature regions of 60 to 100 °C, the  $\sigma$ - $T$  dependence of these two electrolytes show a typical VTF behavior, characteristic of an all-amorphous composition, as already illustrated in other LiX/polymer (e.g., X = Cl, ClO $_4$ , AsF $_6$ , CF $_3$ SO $_3$ ) electrolytes [54,56,57], which is also in accordance with the DSC result that no melting peak is observed at this temperature range (Fig. 2).

Note is that the conductivities of the LiFSI/PEO electrolyte already exceeded  $10^{-3}$  S cm $^{-1}$  at 80 °C, which is higher than those of LiTFSI/PEO electrolyte (EO/Li $^{+}$  = 20 by mole,  $1.0 \times 10^{-3}$  S cm $^{-1}$ , 80 °C), and also higher than other PEO-based SPEs reported in literature, e.g., both LiPF $_6$ /PEO (EO/Li $^{+}$  = 20 by mole,  $\sim 10^{-4}$  S cm $^{-1}$ , 80 °C [12]) and LiBETI/PEO (EO/Li $^{+}$  = 20 by mole,  $6.1 \times 10^{-4}$  S cm $^{-1}$ , 80 °C [26]) electrolytes are lower than  $10^{-3}$  S cm $^{-1}$  at 80 °C. The high ionic conductivities for the LiFSI/PEO electrolyte would be attributed to the higher mobility of the smaller FSI $^{-}$  anion size (i.e., 95 Å $^3$  (FSI $^{-}$ ) vs. 147 Å $^3$  (TFSI $^{-}$ ) [38]), and the highly plasticizing effect of FSI $^{-}$  (i.e., the low fraction of crystalline phase and  $T_g$  as shown in Table 1).

#### 3.4. Lithium-ion transference number

Fig. 5 shows the impedance spectra and time-dependent response of dc polarization for the LiFSI/PEO electrolyte, which was measured on a [Li metal | polymer electrolytes | Li metal] symmetric cell at 80 °C by a combination of complex impedance and potentiostatic polarization method [49–51]. The data of all the measurements, and the calculated values using Eq. (1) for lithium-ion transference number ( $t_{Li^{+}}$ ) for both the LiFSI/PEO and LiTFSI/PEO electrolytes are summarized in Table 2.

As seen from Table 2, the values of lithium-ion transference number ( $t_{Li^{+}}$ ) for these two electrolytes are comparable, and 0.14 for the former and 0.18 for the latter at 80 °C, which is in agreement with those of 0.2 – 0.3 reported for LiTFSI/PEO (EO/Li $^{+}$  = 20 by mole) [24] and 0.16 for LiTFSI/PEO (EO/Li $^{+}$  = 20 by mole) [58], but slightly lower than those of lithium salt/PEO with larger anion size, such as  $t_{Li^{+}}$  = 0.33 for LiBETI/PEO (EO/Li $^{+}$  = 20 by mole) [30]. This would be expected from the smaller anion size for FSI $^{-}$  than for BETI $^{-}$ , which makes the former anion more mobile than the latter one. Nevertheless, the value of  $t_{Li^{+}}$  for LiFSI/PEO is still acceptable in consideration of its high conductivity as shown in Fig. 4.

#### 3.5. Electrochemical stability

The anodic electrochemical stabilities of the LiFSI/PEO and LiTFSI/PEO (EO/Li $^{+}$  = 20 by mole) electrolytes were examined by linear sweep voltammetry. Fig. 6 compares the linear sweep voltammograms of these two electrolytes at 80 °C. As seen from Fig. 6, the LiFSI/PEO (EO/Li $^{+}$  = 20 by mole) electrolyte shows an oxidation potential at about 5.3 V vs. Li $^{+}$ /Li, which is slightly lower than that of 5.7 V for the LiTFSI/PEO (EO/Li $^{+}$  = 20 by mole) one. This would be explained by the concurrent oxidative decomposition of both the FSI $^{-}$  anion and PEO matrix in the high potential regions [59], and the oxidation of FSI $^{-}$  occurred at lower potentials than that of TFSI $^{-}$  during the anodic scan, as already observed in our previous study [36,37], thus made the overall oxidation of LiFSI/PEO electrolyte initiate at a lower potential. However, as seen

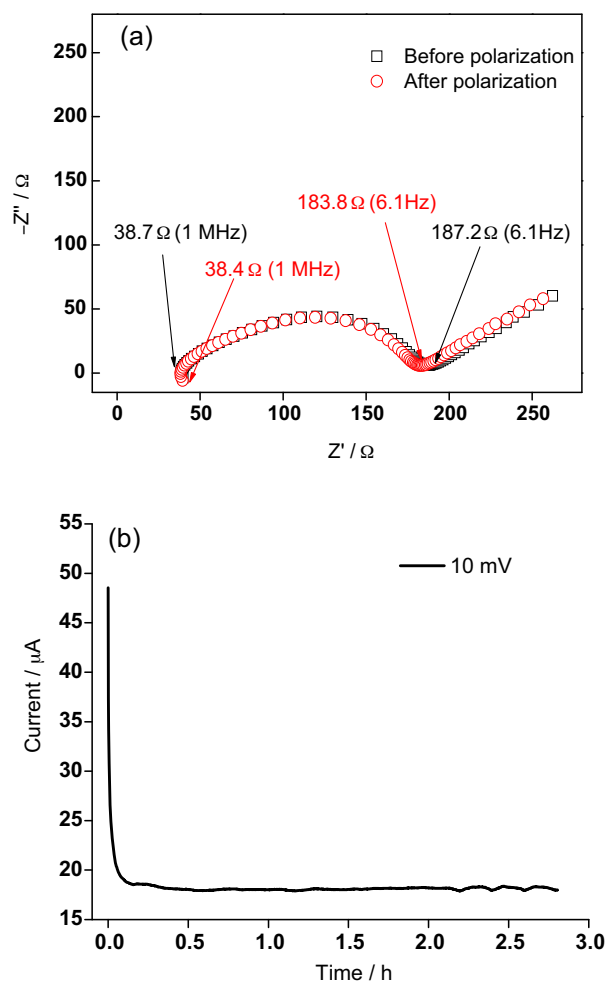


Fig. 5. Impedance spectra (a) and time-dependence response of dc polarization (b) for the LiFSI/PEO (EO/Li $^{+}$  = 20 by mole) electrolyte obtained on the symmetric cell of [Li metal | polymer electrolytes | Li metal] at 80 °C, being polarized with a potential of 10 mV.

in Fig. 6, the LiFSI/PEO electrolyte is still electrochemically stable enough for the use as electrolyte for Li batteries.

Fig. 7 shows the cyclic voltammograms of the LiFSI/PEO and LiTFSI/PEO (EO/Li $^{+}$  = 20 by mole) electrolytes at 80 °C, which was measured on a stainless steel (SS) electrode. As shown in Fig. 7a, the

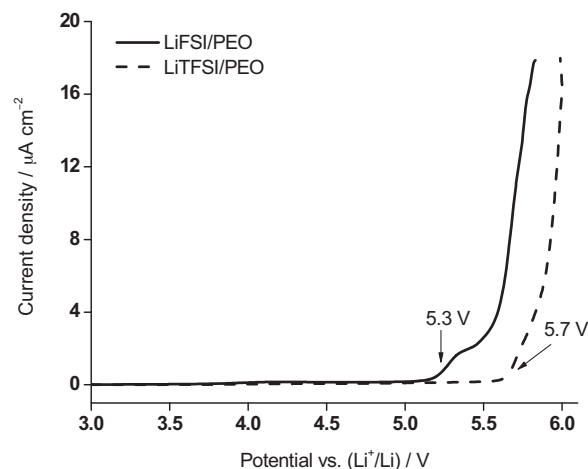


Fig. 6. Linear sweep voltammograms of polymer electrolyte of LiX/PEO (X = FSI and TFSI) with a molar ratio EO/Li $^{+}$  = 20 at 80 °C; scan rate: 0.5 mV s $^{-1}$ .

**Table 2**

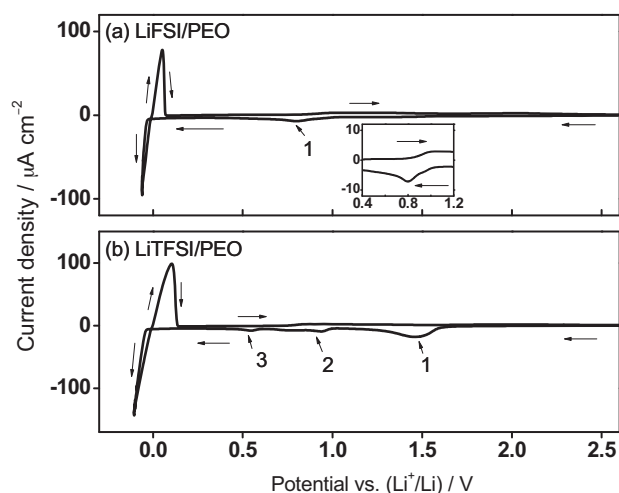
Measured values for the parameters in Eq. (1) and the corresponding calculated values of lithium-ion transference numbers ( $t_{Li^+}$ ) at 80 °C.

| Electrolyte (EO/Li <sup>+</sup> ratio by mole) | $I_0$ (μA) | $I_s$ (μA) | $R_i$ (Ω) | $R_f$ (Ω) | $R_i^0$ (Ω) | $R_i^s$ (Ω) | $\Delta V$ (mV) | $t_{Li^+}$ |
|------------------------------------------------|------------|------------|-----------|-----------|-------------|-------------|-----------------|------------|
| LiFSI/PEO (20)                                 | 48.6       | 18.0       | 38.7      | 38.4      | 148.5       | 145.4       | 10              | 0.14       |
| LiTFSI/PEO (20)                                | 33.3       | 21.1       | 16.9      | 17.0      | 261.5       | 251.9       | 10              | 0.18       |

voltammograms of the LiFSI/PEO electrolyte shows only a cathodic peak at 0.8 V (i.e., the peak labeled as 1 in Fig. 7a), and appears to be reversible as reflected by the anodic peak at 1.0 V. This reversible reaction is probably associated with a redox process involving electrode surface species, such as oxides [51]. However, three identical minor peaks (i.e., the peaks labeled as 1, 2 and 3 in Fig. 7b) are commonly observed for the LiTFSI/PEO (EO/Li<sup>+</sup> = 20 by mole) electrolytes [60]. These peaks might be associated with parasitic processes that could include the reaction of the TFSI<sup>-</sup> anion, the dissolution of Li in the electrode support and/or reaction of impurities [61,62]. In addition, the deposition and dissolution of Li on and from the SS substrate are observed between −0.2 and 0.2 V in both voltammograms, as previously reported on other PEO-based SPEs [60–62], indicating that cathodic stability of these electrolytes is acceptable towards Li metal.

### 3.6. Interfacial behavior of Li metal/polymer electrolyte

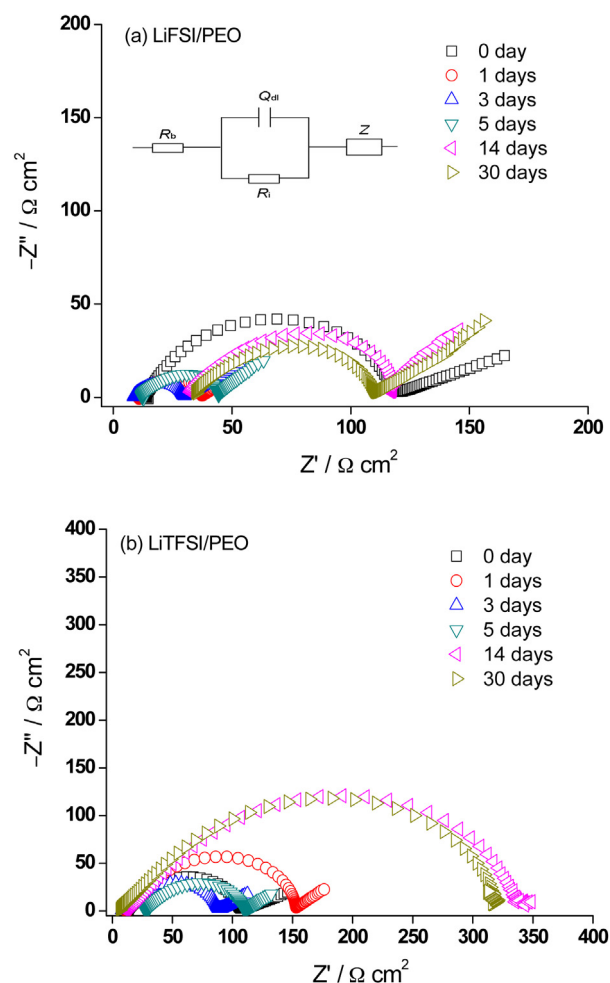
The interface formed between Li metal electrode and SPEs (i.e., solid electrolyte interface, SEI) in Li batteries play a paramount importance in determining the cycling life, rate capability, and safety of the cell, and the properties of the SEI film is highly dependent on the identity of anion, electrolyte preparation procedure and environment, and cell fabrication procedure. The first factor is essentially dictated by the structure of the anion and can be tuned by modifying the molecular structure through understanding the structure-property relationships, while the latter two ones can be controlled through engineering effort. It has been reported that the SEI films formed on the Li metal electrode in the LiBETI/PEO [27] and LiCF<sub>3</sub>SO<sub>3</sub>/PEO [63] electrolytes shows a very good stability both under rest and current flow conditions, while those with LiClO<sub>4</sub>- [64,65], LiBF<sub>4</sub>- [66] and LiTFSI [45,46]-based polymer electrolytes are less stable, and can be significantly enhanced by adding a small amount of dispersed ceramic filler [45,46,64–67]. Therefore, the stability of interface of Li metal electrode with LiFSI/PEO electrolyte, as well as with the representative LiTFSI/PEO electrolyte, is comparatively investigated in this study.



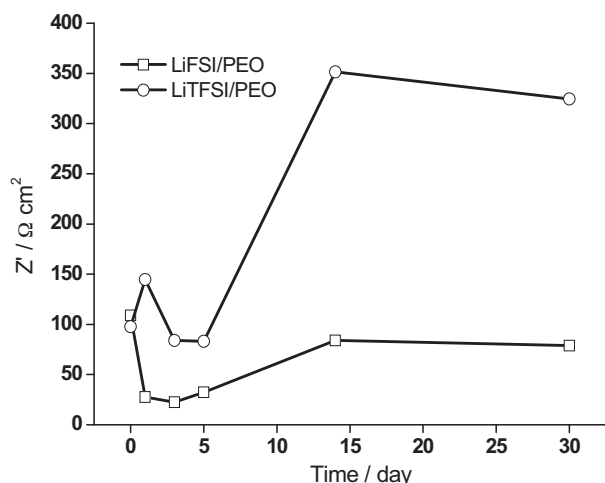
**Fig. 7.** Cyclic voltammograms of polymer electrolyte of LiX/PEO (X = FSI and TFSI) with a molar ratio EO/Li<sup>+</sup> = 20 at 80 °C; scan rate: 0.5 mV s<sup>−1</sup>.

The interfacial stability of Li electrode with both the LiFSI/PEO and LiTFSI/PEO (EO/Li<sup>+</sup> = 20 by mole) electrolytes was investigated by monitoring the impedance evolution of a symmetrical Li | polymer electrolytes | Li cell at 80 °C in rest conditions, as well described by Scrosati and Passerini [27,32,45,63,68]. Fig. 8 shows the impedance spectra for the symmetric cells based on these two electrolytes at 80 °C after various storage times in rest conditions. A simplified equivalent circuit adopted to fit the collected EIS data is shown as inset of Fig. 8a [27]. In this circuit,  $R_b$  is the bulk resistance of polymer electrolyte;  $R_i$  is the Li/SPEs interface resistance;  $Q_{dl}$  is the double layer capacity of Li/SPEs interface;  $Z$  is the diffusive impedance [27].

Fig. 9 compares the values of interfacial resistance of Li metal electrode/electrolyte in both the LiFSI/PEO and LiTFSI/PEO (EO/Li<sup>+</sup> = 20 by mole) electrolytes as a function of storage time at 80 °C, which is obtained by best-fitting the data of the impedance measurements to the equivalent circuit in Fig. 8. As seen from Fig. 9, the value of  $R_i$  of the freshly assembled symmetrical cell for the LiFSI/PEO electrolyte is 109 Ω cm<sup>2</sup>, which is slightly higher than



**Fig. 8.** Normalized impedance spectra for the polymer electrolyte of LiX/PEO (X = FSI and TFSI) with a molar ratio EO/Li<sup>+</sup> = 20, obtained on the symmetric cell of [Li metal | polymer electrolytes | Li metal] at 80 °C.

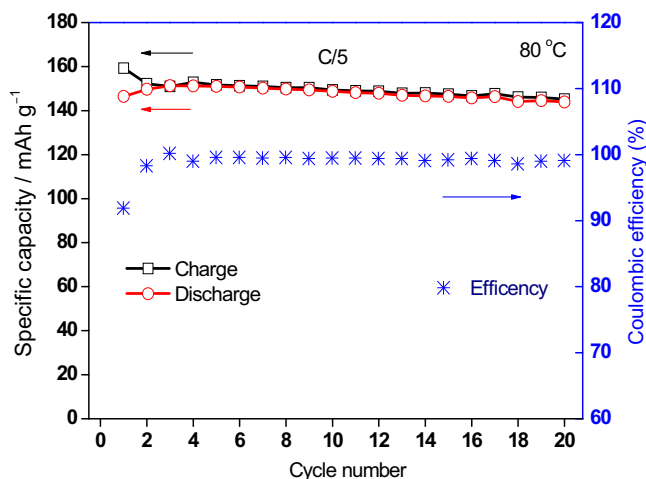


**Fig. 9.** Time evolution of the interfacial resistance ( $R_i$ ) of Li/SPEs after various storage time at 80 °C. Data were obtained by nonlinear fit of raw impedance measurements.

that of 98  $\Omega \text{ cm}^2$  for the LiTFSI/PEO one. After storage for a few days (5 day), the values of  $R_i$  for these two polymer electrolytes gradually decrease to a more or less extent. For instances, the values of  $R_i$  decrease from 109 to 33  $\Omega \text{ cm}^2$  for LiFSI/PEO electrolyte, and from 98 to 83  $\Omega \text{ cm}^2$  for LiTFSI/PEO electrolyte at day 5, indicating the formation and recombination of the interface of Li metal electrode/electrolyte was occurring [27]. With further increasing the storage time, the values of  $R_i$  for the LiFSI/PEO electrolyte remain relatively stable at the value of  $< 90 \Omega \text{ cm}^2$ , e.g.,  $R_i = 84 \Omega \text{ cm}^2$  at day 14 and 79  $\Omega \text{ cm}^2$  at day 30. These  $R_i$  values for the LiFSI/PEO electrolyte at 80 °C are comparable to those for the excellent LiBETI/PEO electrolyte (35–65  $\Omega \text{ cm}^2$  [27]) at 75 °C. This suggests that the LiFSI/PEO electrolyte has excellent compatibility with Li metal, especially at the high temperature (80 °C) required for the operation of Li battery using SPEs. By contrast, the values of  $R_i$  of the LiTFSI/PEO electrolyte substantially increase from 83  $\Omega \text{ cm}^2$  at day 5 to 352  $\Omega \text{ cm}^2$  at day 30, indicating that a stable interface between the Li metal and LiTFSI/PEO electrolyte has not achieved. This unstable interfacial behavior of the LiTFSI/PEO electrolyte with Li metal was also observed in literature [45,46,68,69], and was improved by adding a small amount of dispersed ceramic filler [45]. The above results suggest that the interface formed between the Li metal electrode and SPEs is more stable in the LiFSI/PEO electrolyte than in the LiTFSI/PEO one. Although the mechanism for forming the stable interface between Li metal and LiFSI/PEO electrolyte is not clear presently, it seems that inorganic  $\text{FSO}_2^-$  group in LiFSI play a critical role in improving the stability of interface of Li/SPEs, and similar phenomena were also observed in the  $\text{FSO}_2^-$  group based lithium salts in conventional liquid carbonate electrolytes [43,44] and ionic liquids [36].

### 3.7. Li/LiFePO<sub>4</sub> battery

Fig. 10 shows cycling performances (charge/discharge capacity and coulombic efficiency (CE) vs. cycle number) of the Li/LiFePO<sub>4</sub> cell using dry LiFSI/PEO electrolyte at 80 °C. This cell could deliver a high specific charge capacity of 159  $\text{mAh g}^{-1}$  and discharge capacity of 146  $\text{mAh g}^{-1}$  at the first cycle at C/5 rate (1 C corresponding to 0.5 mA, theoretically 170  $\text{mAh g}^{-1}$  for LiFePO<sub>4</sub>). The CE of the first cycle is 91% and reaches ca. 99% immediately after the first cycle, suggesting relatively stable interfacial layers on both the Li metal anode and LiFePO<sub>4</sub> cathode have been formed at the first cycle. The discharge capacity of the LiFSI/PEO-based Li/LiFePO<sub>4</sub> cell is 144



**Fig. 10.** Charge-discharge capacity and coulombic efficiency vs. cycle number for Li/LiFePO<sub>4</sub> cells using LiFSI/PEO (EO/Li<sup>+</sup> = 20 by mole) electrolyte at 80 °C. Charge/discharge rate: 0.2/0.2 C. Color: black (charge), red (discharge), and blue (coulombic efficiency).

$\text{mAh g}^{-1}$  at the 20th cycle, indicating that LiFSI/PEO electrolyte can afford a good cycling performance. This would be attributable to the high conductivity (Fig. 4a) and good Li/electrolyte interface (Fig. 9) in the LiFSI/PEO electrolyte.

## 4. Conclusion

A new type of solid polymer electrolyte, LiFSI/PEO, has been prepared and characterized. LiFSI can significantly reduce the crystallinity and melting temperature of neat PEO due to the highly plasticizing effect of the FSI<sup>−</sup> anion. The polymer electrolyte of LiFSI/PEO exhibits ionic conductivity as high as  $1.3 \times 10^{-3} \text{ S cm}^{-1}$  at 80 °C. The interface of Li metal electrode/electrolyte formed in the LiFSI/PEO electrolyte shows a very good stability at 80 °C. Li/LiFePO<sub>4</sub> cell using dry LiFSI/PEO electrolyte shows excellent cycling performance at 80 °C. Thus, LiFSI would be a promising salt of polymer electrolyte for Li batteries.

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